

Osteogenic Stimulators

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Tracking Information

Publication Number

100-3

Manual Section Number

150.2

Manual Section Title

Osteogenic Stimulators

Version Number

2

Effective Date of this Version

04/27/2005

Implementation Date

08/01/2005

Description Information

Benefit Category

Durable Medical Equipment

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Electrical Osteogenic Stimulators

A. General

Electrical stimulation to augment bone repair can be attained either invasively or non-invasively. Invasive devices provide electrical stimulation directly at the fracture site either through percutaneously placed cathodes or by implantation of a coiled cathode wire into the fracture site. The power pack for the latter device is implanted into soft tissue near the fracture site and subcutaneously connected to the cathode, creating a self-contained system with no external components. The power supply for the former device is externally placed and the leads connected to the inserted cathodes. With the non-invasive device, opposing

pads, wired to an external power supply, are placed over the cast. An electromagnetic field is created between the pads at the fracture site.

Ultrasonic Osteogenic Stimulators

A. General

An ultrasonic osteogenic stimulator is a noninvasive device that emits low intensity, pulsed ultrasound. The device is applied to the surface of the skin at the fracture site and ultrasound waves are emitted via a conductive coupling gel to stimulate fracture healing. The ultrasonic osteogenic stimulators are not be used concurrently with other non-invasive osteogenic devices.

Indications and Limitations of Coverage

Electrical Osteogenic Stimulators

B. Nationally Covered Indications

1. Noninvasive Stimulator

The noninvasive stimulator device is covered only for the following indications:

- Nonunion of long bone fractures;
- Failed fusion, where a minimum of 9 months has elapsed since the last surgery;
- Congenital pseudarthroses;
- Effective July 1, 1996, as an adjunct to spinal fusion surgery for patients at high risk of pseudarthrosis due to previously failed spinal fusion at the same site or for those undergoing multiple level fusion. A multiple level fusion involves 3 or more vertebrae (e.g., L3-L5, L4-S1, etc).
- Effective September 15, 1980, nonunion of long bone fractures is considered to exist only after 6 or more months have elapsed without healing of the fracture.
- Effective April 1, 2000, nonunion of long bone fractures is considered to exist only when serial radiographs have confirmed that fracture healing has ceased for 3 or more months prior to starting treatment with the electrical osteogenic stimulator. Serial radiographs must include a minimum of 2 sets of radiographs, each including multiple views of the fracture site, separated by a minimum of 90 days.

2. Invasive (Implantable) Stimulator

The invasive stimulator device is covered only for the following indications:

- Nonunion of long bone fractures;
- Effective July 1, 1996, as an adjunct to spinal fusion surgery for patients at high risk of pseudarthrosis due to previously failed spinal fusion at the same site or for those undergoing multiple level fusion. A multiple level fusion involves 3 or more vertebrae (e.g., L3-L5, L4-S1, etc).
- Effective September 15, 1980, nonunion of long bone fractures is considered to exist only after 6 or more months have elapsed without healing of the fracture.

- Effective April 1, 2000, nonunion of long bone fractures is considered to exist only when serial radiographs have confirmed that fracture healing has ceased for 3 or more months prior to starting treatment with the electrical osteogenic stimulator. Serial radiographs must include a minimum of 2 sets of radiographs, each including multiple views of the fracture site, separated by a minimum of 90 days.

Ultrasonic Osteogenic Stimulators

B. Nationally Covered Indications

Effective January 1, 2001, ultrasonic osteogenic stimulators are covered as medically reasonable and necessary for the treatment of nonunion fractures. In demonstrating non-union fractures, CMS expects:

- A minimum of 2 sets of radiographs, obtained prior to starting treatment with the osteogenic stimulator, separated by a minimum of 90 days. Each radiograph set must include multiple views of the fracture site accompanied with a written interpretation by a physician stating that there has been no clinically significant evidence of fracture healing between the 2 sets of radiographs; and,
- Indications that the patient failed at least one surgical intervention for the treatment of the fracture.
- Effective April 27, 2005, upon reconsideration of ultrasound stimulation for nonunion fracture healing, CMS determines that the evidence is adequate to conclude that noninvasive ultrasound stimulation for the treatment of nonunion bone fractures prior to surgical intervention is reasonable and necessary. In demonstrating non-union fractures, CMS expects:
 - A minimum of 2 sets of radiographs, obtained prior to starting treatment with the osteogenic stimulator, separated by a minimum of 90 days. Each radiograph set must include multiple views of the fracture site accompanied with a written interpretation by a physician stating that there has been no clinically significant evidence of fracture healing between the 2 sets of radiographs.

C. Nationally Non-Covered Indications

Nonunion fractures of the skull, vertebrae and those that are tumor-related are excluded from coverage.

Ultrasonic osteogenic stimulators may not be used concurrently with other non-invasive osteogenic devices.

Ultrasonic osteogenic stimulators for fresh fractures and delayed unions remains non-covered.

(This NCD last reviewed June 2005.)

Claims Processing Instructions

[TN 597 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

41

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R41NCD.pdf> 

Revision History

06/2005 - Determined evidence adequate to conclude that non-invasive ultrasound stimulation for treatment of nonunion bone fractures prior to surgical intervention is reasonable and necessary. Effective date 04/27/2005. Implementation date 08/01/2005. ([TN 41](#) ) (CR 3836)

11/2000 - Permitted coverage for ultrasonic osteogenic stimulators when two sets of radiographs document nonunion of fracture, and patient has failed at least one surgical intervention for treatment of fracture. Effective and implementation dates 01/01/2001. ([TN 131](#) )

02/2000 - Defined nonunion of long bone fractures when fracture healing has ceased for 3 or more months as confirmed by serial radiographs. Effective and implementation dates 04/01/2000. ([TN 123](#) ) (CR 1085)

03/1997 - Denied coverage for osteogenic stimulators due to insufficient evidence to support medical necessity. Effective date 04/03/1997. (TN 92)

06/1996 - Expanded coverage to include use as an adjunct to spinal fusion surgery and provided clarification when noninvasive osteogenic stimulation is indicated after failed fusion. Effective date 07/01/1996. (TN 85)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Ultrasound Stimulation for Nonunion Fracture Healing \(CAG-00022N\)](#) 
- [First reconsideration for Ultrasound Stimulation for Nonunion Fracture Healing \(CAG-00022R\)](#) 
- [Original Consideration for Electrical Stimulation for Fracture Healing \(CAG-00043N\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Osteogenic Stimulators	2	04/27/2005 - N/A	You are here
Osteogenic Stimulation	1	01/01/2001 - 04/27/2005	View